COST-AWARE HARNESS OPTIMIZATION



PE6203 CA2
TUA-Bench terminal-agent evaluation

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Experimental setup

40 eval tasks

8 tasks per family
10 separate dev tasks
Frozen subset and oracle gate
One run per configuration

Role	Model	Provider
Base	GPT-OSS-20b	Darkbloom
Expensive	GPT-OSS-120b	Venice

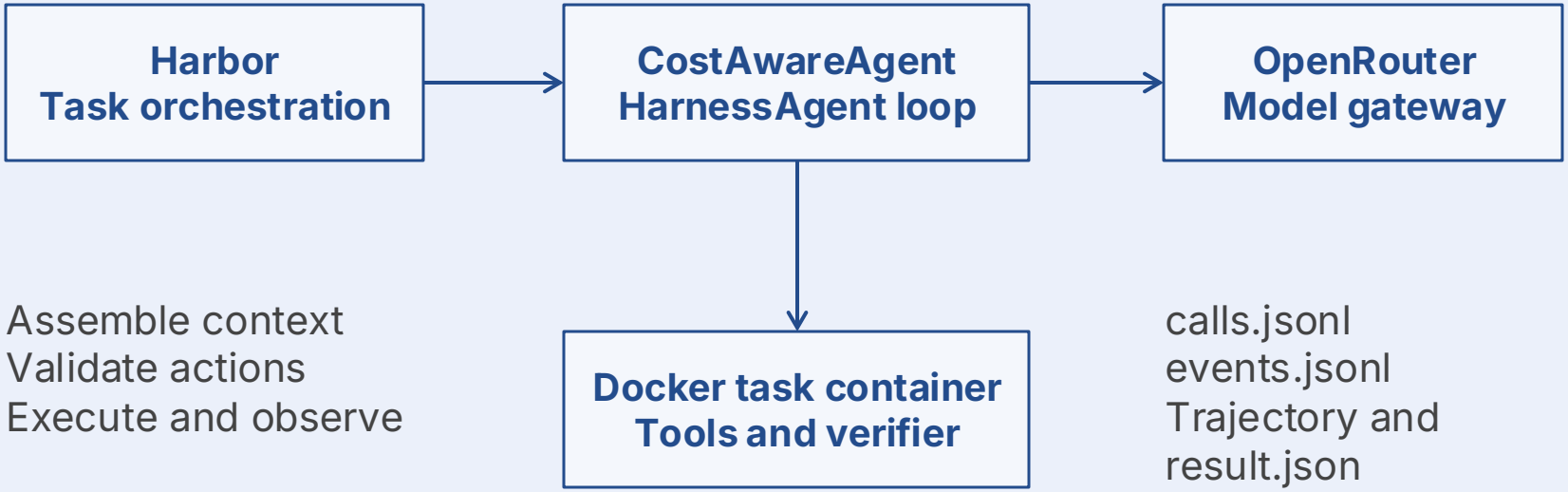
Fixed routing, fallback disabled
Low reasoning, max_tokens 8192

438 / 440 trials scored

11 configurations

Task: 2,400 s. C3/V: 50 steps, $0.01.
V-E: 50 steps, $0.0167.

Harness architecture and Harbor integration



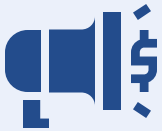
H0: loop only. V: full harness. V-E: full harness, 120b.

C0, C1 and C3



C0 PROMPT

Task-specific instructions
Explicit tool-use protocol
Clear submission command



C1 CONTEXT

Offload long outputs
Shorten observations
Compact older context



C3 BUDGET

50-step allowance
$0.01 per task
Budget status in context

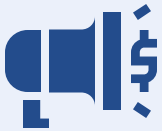
C3 changes command timeout from 30 s to 180 s. V-E uses a $0.0167 cap.

C2, C4 and C5



C2 TOOLS

Read, write, edit and search
Validate actions
Five format-error allowance



C4 RECOVERY

Detect repeated commands
Check stalled observations
Add recovery nudges



C5 SUBMIT GATE

Check required artifacts
Flag missing outputs
Verify before submission

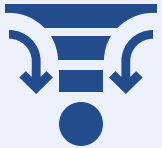
All components are independently switchable. Deterministic checks add no model calls.

Guardrails and false-positive review



ISOLATION

Task work stays inside the Docker container



INTEGRITY

Always-on rules restrict grader paths and destructive actions



COMMAND CHECKS

C2 validates syntax, interactive commands and submission

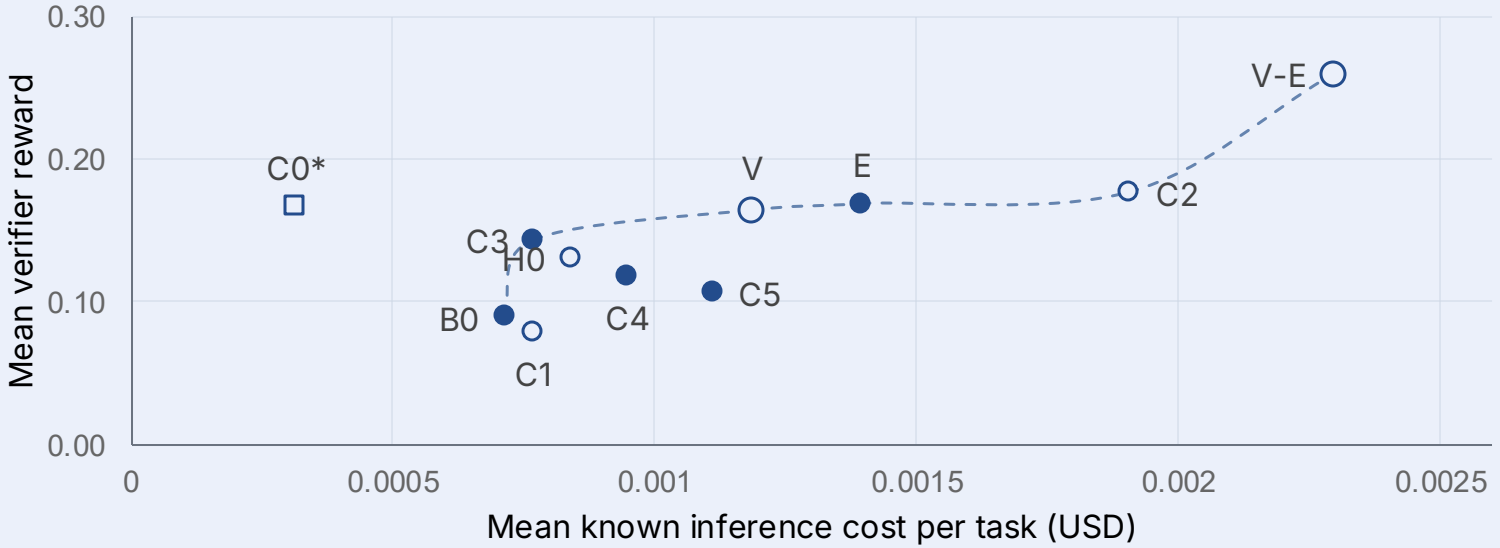


FALSE POSITIVES

sudo becomes log-only U-INT-05
Heredoc defects fixed after eval

Reported evaluation: harness-v1. Later fixes: harness-v1.1. Results remain unchanged.

Accuracy and cost



Hollow = lower-bound cost. Dashed frontier admits lower bounds. C0*: 38/40 scored, excluded.

Strict frontier: B0, C3, E. C0–C5 = H0 + component.
Reward includes partial credit.

Baseline comparison and cost breakdown

Config	Reward	Input / output tokens	Calls	Cost per run
B0	0.089	1.19 M / 79 k	449	$0.0286
H0	0.131	1.44 M / 84 k	496	≥ $0.0336
V	0.164	2.22 M / 84 k	659	≥ $0.0474
E	0.169	1.56 M / 60 k	514	$0.0558
V-E	0.260	2.71 M / 71 k	795	≥ $0.0919

Harness-made model calls: $0

Estimated injected prompt cost: V $0.0061 (13%) V-E $0.0126 (14%)

One run = 40 tasks. ≥ indicates a lower bound. Verifier charges excluded. Format feedback is separate.

Component ablation against H0

Component	Reward	Δ vs H0	95% interval	Cost per run
C0 Prompt*	0.166	+0.046	+0.000 to +0.110	≥ $0.0126*
C1 Context	0.079	−0.052	−0.143 to +0.029	≥ $0.0307
C2 Tools	0.176	+0.046	−0.046 to +0.137	≥ $0.0763
C3 Budget	0.143	+0.013	−0.041 to +0.079	$0.0307
C4 Recovery	0.118	−0.012	−0.092 to +0.063	$0.0379
C5 Submit gate	0.106	−0.024	−0.129 to +0.083	$0.0444

*C0: 38 paired tasks, H0 mean 0.120 there. Calls: 247 vs 470. Other rows: 40 tasks.

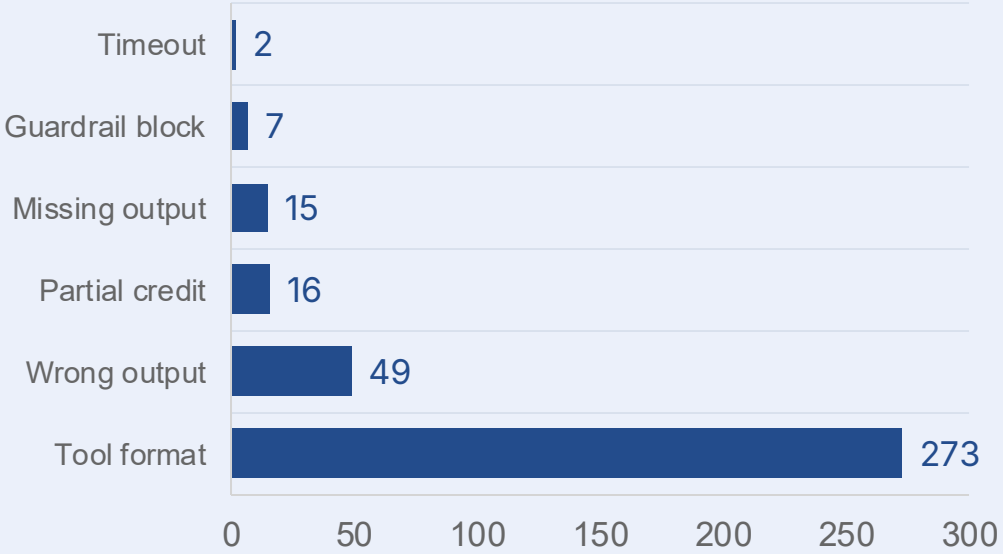
C0 is borderline significant (lower CI endpoint +0.000098); other intervals span zero.

Failure classification

273

Unparsable tool calls
out of 438 scored trials

297 harness-level
65 model-level
45 unclassified
31 full success



Partial credit belongs to the model-level group. Archived infrastructure attempts are excluded.

Representative failure trajectories

HARNESS-LEVEL

B0 / task 040

Three plain-text replies
No executable tool call
RepeatedFormatError

MODEL-LEVEL

V-E / task 115

Writes [] to keybindings.json
Missing “-list.find” rule
Valid execution, wrong result

HARNESS-INDUCED

V / task 071

Valid heredoc rejected 5×
U-FMT-03 false positive
Validation blocks progress

Audit: all 13 U-FMT-03 blocks were false positives. Later fixes in harness-v1.1 do not change the v1 evaluation.

Harness gains with a stronger model

20b: V – B0

+0.075

95% CI: +0.008 to +0.158

120b: V-E – E

+0.091

95% CI: −0.013 to +0.203

No observed shrinkage, with wide and overlapping intervals

C0 and C2 show the main positive signals. V gains +0.033 over H0, suggesting overlapping benefits.



CONCLUSIONS

V improves the cheap-model baseline
Clear prompts provide the clearest gain
Controls need their own evaluation

Single run / 40 tasks / lower-bound costs
Two infrastructure failures / v1 results retained

Thank you. Questions?